Point-wise mutual information (PMI) :-

PMI score tells us how much more/less likely the co-occurrence is than if the words were independent.

$$PPMI(W,C) = \log_2 \frac{p(W,C)}{p(W) \quad p(C)}$$

Positive Point-wise mutual information (PPMI):-

PMI score could range from −∞ to + ∞

But the negative values are problematic

- Things are co-occurring less than we expect by chance
- Unreliable without enormous corpora
  - Imagine w1 and w2 whose probability is each 10-6
  - Hard to be sure p(w1,w2) is significantly different than 10-12
  - Plus it’s not clear people are good at “un-relatedness”

So negative PMI score tells us that two words co-occur less than we expect. Because for infrequent words we do not have enough data to accurately determine negative PMI values .So to handle this problem we just replace negative PMI values by 0.

Positive PMI (PPMI) between word1 and word2 can be written as follows-

$$PPMI(Word1, Word2) = max(\log_2 \frac{p(Word1, \quad Word2)}{p(Word1) \quad p(Word2)}, 0)$$

Above equation can be rewritten as follows-

$$\begin{Bmatrix} \text{PMI(W, C)} & , if \text{ PMI(W, C)} > 0 \\ 0 & , if \text{ PMI(W, C)} < 0 \end{Bmatrix}$$

Where,

$$p(W, C) = \frac{f_{ij}}{\sum_{i=1}^{W} \sum_{j=1}^{C} f_{ij}}$$

$$p(W_i) = \frac{\sum_{j=1}^{C} f_{ij}}{N}$$

$$p(C_j) = \frac{\sum_{i=1}^{W} f_{ij}}{N}$$

Where ,

- p(W,C) -is the probability of seeing Target word w and context word c together.
- p(W) & p(C) – the probability of occurring Target word w & context word C , if they're independent
- f_{ij} is number of times W_i occurs in context C_j

Computing PPMI on a term-context matrix:-

Let's understand the PPMI with below matrix representation with W rows (words) and C columns (contexts, e.g. in the form **of Words.)**

Count(W, Context)					
	Computer	Data	Pinch	Result	Sugar
Apricot	0	0	1	0	1
Pineapple	0	0	1	0	1
Digital	2	1	0	1	0
Information	1	6	0	4	0

P(W=information , C=data) =6/19=0.32

P(W=information) = 11/19 = 0.58

P(C= data) = 7/19 = 0.37

P(W, Context)						
	Computer	Data	Pinch	Result	Sugar	P(W)
Apricot	0	0	1	0	1	0.11
Pineapple	0	0	1	0	1	0.11
Digital	2	1	0	1	0	0.21
Information	1	6	0	4	0	0.58
P(Context)	0.16	0.37	0.11	0.26	0.11	

So from above information let's calculate PMI score for word "information" getting co-occurred with context word "data"-

$$\text{PMI}(W, C) = \log \frac{p(W,C)}{p(W)p(C)}$$

$$\text{PMI (information, data)} = \log \frac{0.32}{0.37*0.58} = \sim 0.58$$

PMI (information, data)=0.58

Similarly calculate for rest of the word and let's populate those scores in matrix like below- Remember,

$$\begin{Bmatrix} \text{PMI}(W, C) & , if\ \text{PMI}(W, C) > 0 \\ 0 & , if\ \text{PMI}(W, C) < 0 \end{Bmatrix}$$

PPMI(W, Context)					
	Computer	Data	Pinch	Result	Sugar
Apricot	-	-	2.25	-	2.25
Pineapple	-	-	2.25	-	2.25
Digital	1.66	0	-	0	-
Information	0	0.57	-	0.47	-

Weighting PMI:-

If you notice from above matrix then you'll know that PMI is biased toward infrequent events. Very rare words have very high PMI values . So we can improve PMI further with two possible solutions-

- Use add-k smoothing
- Give rare words slightly higher probabilities (which has a similar effect)

K-Smoothing in PMI computation:-

As we've seen PMI is biased toward infrequent events , in our case possibility of two words getting co-occurred together.

So we add 2 in every cell of co-occurrence matrix like below-

K-Smoothing in PMI computation

Add 2 smoothed Count(W, Context)					
	Computer	Data	Pinch	Result	Sugar
Apricot	2	2	3	2	3
Pineapple	2	2	3	2	3
Digital	4	3	2	3	2
Information	3	8	2	6	2

P(W, Context)						
	Computer	Data	Pinch	Result	Sugar	P(W)
Apricot	0.03	0.03	0.05	0.03	0.05	0.20
Pineapple	0.03	0.03	0.05	0.03	0.05	0.20
Digital	0.07	0.05	0.03	0.05	0.03	0.24
Information	0.05	0.14	0.03	0.10	0.03	0.36
P(Context)	0.19	0.25	0.17	0.22	0.17	

K-Smoothing in PMI computation

Add 2 smoothed Count(W, Context)					
	Computer	Data	Pinch	Result	Sugar
Apricot	2	2	3	2	3
Pineapple	2	2	3	2	3
Digital	4	3	2	3	2
Information	3	8	2	6	2

P(W, Context)						
	Computer	Data	Pinch	Result	Sugar	P(W)
Apricot	0.03	0.03	0.05	0.03	0.05	0.20
Pineapple	0.03	0.03	0.05	0.03	0.05	0.20
Digital	0.07	0.05	0.03	0.05	0.03	0.24
Information	0.05	0.14	0.03	0.10	0.03	0.36
P(Context)	0.19	0.25	0.17	0.22	0.17	

PPMI(W, Context)[add-2]					
	Computer	Data	Pinch	Result	Sugar
Apricot	0	0	0.56	0	0.56
Pineapple	0	0	0.56	0	0.56
Digital	0.62	0	0	0	0
Information	0	0.58	0	0.37	0

Weighting PMI: Giving rare context words slightly higher probability

- Raise the context probabilities to $\alpha = 0.75$:

$$\mathrm{PPMI}_\alpha(w,c) = \max(\log_2 \frac{P(w,c)}{P(w)P_\alpha(c)}, 0)$$

$$P_\alpha(c) = \frac{count(c)^\alpha}{\sum_c count(c)^\alpha}$$

- This helps because $P_\alpha(c) > P(c)$ for rare c
- Consider two events, P(a) = .99 and P(b)=.01 (here we use probability to show the effect)
- $P_\alpha(a) = \frac{.99^{.75}}{.99^{.75}+.01^{.75}} = .97 \quad P_\alpha(b) = \frac{.01^{.75}}{.99^{.75}+.01^{.75}} = .03$

PPMI V/S k- smoother PMI:-

PPMI vs K-Smoothing

PPMI(W, Context)					
	Computer	Data	Pinch	Result	Sugar
Apricot	-	-	2.25	-	2.25
Pineapple	-	-	2.25	-	2.25
Digital	1.66	0	-	0	-
Information	0	0.57	-	0.47	-

PPMI(W, Context) [Add 2]					
	Computer	Data	Pinch	Result	Sugar
Apricot	0	0	0.56	0	0.56
Pineapple	0	0	0.56	0	0.56
Digital	0.62	0	0	0	0
Information	0	0.58	0	0.37	0